# Half-degree irreducibles over $\mathbb{F}_2$: what is proved and what remains open

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**Status.** This note does not prove Lemire's conjecture. The estimate shown in red in Figure 1 is still open.

**Abstract**

Barrett reduction over $\mathbb{F}_2$ is especially cheap when an irreducible polynomial $f$ of degree $n$ satisfies $\deg(f - x^n) \le \lfloor n/2 \rfloor$. Daniel Lemire asked whether such a polynomial exists in every degree. We translate the condition into a question about one ray class and then count that class with the polynomial von Mangoldt function. Most of the resulting character sum can now be bounded. The part we cannot bound is a signed sum over characters of high 2-power order. We state the needed bound and prove that it would finish the argument. Separate Separate certified tests cover every degree through 3000.

## 1 The argument

Put $\ell = \lceil n/2 \rceil - 1$. Figure 1 shows what is proved and what is not.

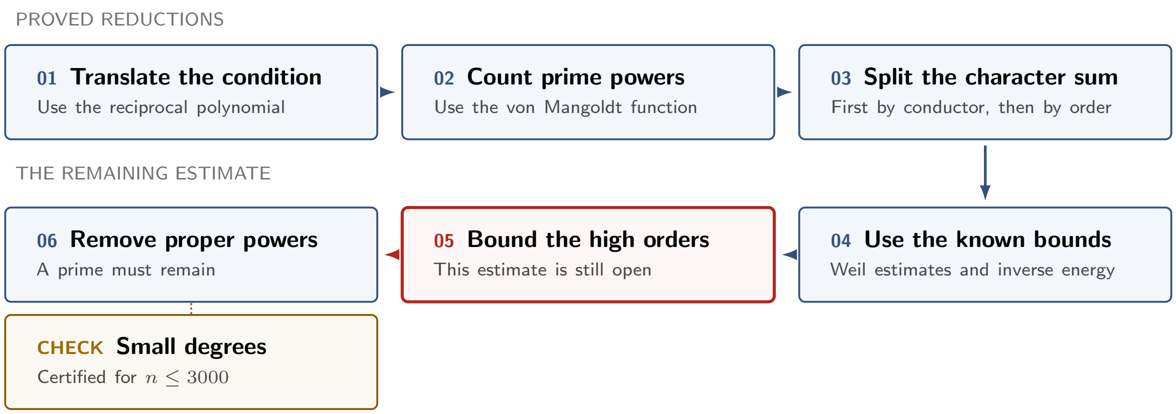


Figure 1: Steps 1–4 and 6 are proved. Step 5 is open. The gold box records the separate computation for $n \le 3000$.

## 2 What we can prove

**1. Translate the condition (PROVED).** For $j \ge 0$ set

$$\mathcal{E}_j = (1 + x\mathbb{F}_2[x])/(x^{j+1}), \qquad |\mathcal{E}_j| = 2^j,$$

and, for monic $F$, let $\langle F \rangle_j = x^{\deg F} F(x^{-1}) \pmod{x^{j+1}}$. Then $\deg(f - x^n) \le \lfloor n/2 \rfloor$ if and only if $\langle f \rangle_\ell = 1$. So the conjecture asks whether this ray class contains a prime of degree $n$. Reversing $f$ gives an equivalent arithmetic-progression form: a prime $P$ of degree $n$ with $P \equiv 1 \pmod{x^{\lceil n/2 \rceil}}$, i.e. Linnik's problem with exponent exactly 2 for the residue 1 and the prime-power modulus $x^{\lceil n/2 \rceil}$. In Witt-vector terms (Katz's splitting $\mathcal{E}_j \cong \prod_{k \text{ odd} \le j} \mathbb{Z}/2^{e_k}$, $e_k = \lfloor \log_2(j/k) \rfloor + 1$), the class of $\alpha \in \mathbb{F}_{2^n}$ is the vector of Galois-ring traces $\mathrm{Tr}_{\mathrm{GR}(2^{e_k},n)}(\tilde{\alpha}^k) \bmod 2^{e_k}$, $\tilde{\alpha}$ the Teichmüller lift; the identity class is the set of $\alpha$ all of whose odd-power traces vanish to these dyadic precisions.

**2. Count prime powers (PROVED).** Define

$$N_j(b) = \sum_{\substack{F \text{ monic, } \deg F = n \\ \langle F \rangle_j = b}} \Lambda(F).$$

Logarithmic differentiation gives this weighted count exactly. At the odd endpoint $n = 2\ell + 1$, $N_\ell(1) = 1 + nI_n(1)$. At the even endpoint $n = 2\ell + 2$, odd proper powers do not occur in the identity class. A direct count bounds the squares and the other even powers. In either case, it is enough to prove

$$N_\ell(1) > 2^{n-\ell} - 2^\ell. \tag{1}$$

**3. Split the count by conductor (PROVED).** Each map $\mathcal{E}_j \to \mathcal{E}_{j-1}$ has two-element fibres. If $H_j(b)$ is the difference of the two child populations above $b$, repeated binary splitting gives

$$2^\ell N_\ell(e) = 2^n + \sum_{j=1}^{\ell} \varepsilon_j(e) 2^{j-1} H_j(b_j(e)), \qquad \varepsilon_j(e) \in \{\pm 1\}. \tag{2}$$

For primitive characters $X_j$ of conductor $j$,

$$H_j(b) = 2^{1-j} \sum_{\chi \in X_j} \overline{\chi(b)} S_n(\chi), \quad |H_j(b)| \le (j-1)2^{\lceil n/2 \rceil},$$

The function-field Riemann hypothesis gives the inequality on the right. It is strong enough for the small conductors. If we use it separately at every conductor, however, the final bound is too large by a factor of order $\ell$.

**4. Prove the bilinear bound (PROVED, but not enough).** The needed inverse-energy bound holds in characteristic two, including the case where products wrap modulo $x^r$. The proof groups collisions by their $x$-adic valuation. A Padé approximation turns each group into a problem about factoring polynomials of bounded degree. Lift counts and divisor bounds then finish the estimate. This proves the bilinear input used in Bagshaw's Type-I/II argument. It does not finish our problem, because applying the bound one term at a time destroys the signs in (2).

**5. Add the large conductors before taking absolute values (PROVED).** Let $c = \lceil \log_2 \ell \rceil$, $a = \ell - c - 1$, and

$$W_{\ell,n} = 2^{\lceil n/2 \rceil} \sum_{1 \le j < a} (j-1)2^{j-1}, \quad B_{\ell,n} = 2^{2\ell} - W_{\ell,n} > 0.$$

For the identity class, these terms telescope:

$$C_{\ell,n} := \sum_{j=a}^{\ell} 2^{j-1} H_j(1) = 2^\ell N_\ell(1) - 2^{a-1} N_{a-1}(1). \tag{3}$$

It would now be enough to prove

$$C_{\ell,n} > -B_{\ell,n} \tag{REL}$$

The smaller conductors contribute at least $-W$. With (REL), this gives $2^\ell N_\ell(1) - 2^n > -W - B = -2^{2\ell}$, which is (1).

**6. Split the remaining characters by order (PROVED).** For $s \ge 0$, write

$$\mathcal{H}_{j,s} = \{\chi : \chi^{2^s} = 1\}, \quad h_{j,s} = 2^{j - \lfloor j/2^s \rfloor}, \quad P_{j,s} = \sum_{g \in 2^s \mathcal{E}_j} N_j(g).$$

Finite-group orthogonality gives the sum over characters of conductor $j$ and exact order $2^s$:

$$T_{j,s} = h_{j,s}P_{j,s} - h_{j,s-1}P_{j,s-1} - h_{j-1,s}P_{j-1,s} + h_{j-1,s-1}P_{j-1,s-1}.$$

Let $Q$ be the largest power of two with $3Q\lceil \log_2 \ell \rceil \le \ell$. The Weil bound handles every order at most $Q$. This leaves $O(\log\log \ell)$ high orders at each conductor. When $\ell = 200$, it handles 20 of the 67 conductor–order pairs and leaves 47.

# 3 What remains open

**7. Bound the high orders (OPEN).** Suppose $a \le j \le \ell$, $2^s > Q$, and the corresponding set of characters is nonempty. The following bound would be enough:

$$4\ell\, |T_{j,s}(n)| \le \#X_{j,s}(j-1)2^{\lceil n/2 \rceil}. \tag{HWO}$$

We can also state the same bound without characters. Put $\Delta_{j,s} = 2P_{j,s} - P_{j-1,s}$, $d_s = \lfloor (j-1)/2^s \rfloor$, and $R_{j,s} = 2^{d_{s-1} - d_s}$. There are two cases. If $2^{s-1} \nmid j$, (HWO) is equivalent to

$$4\ell\, |R_{j,s}\Delta_{j,s} - \Delta_{j,s-1}| \le (R_{j,s} - 1)(j-1)2^{\lceil n/2 \rceil}. \tag{NSD}$$

If $2^{s-1} \mid j$ but $2^s \nmid j$, the layer is resonant instead, and (HWO) is equivalent to the direct sparse-discrepancy bound

$$4\ell\,|\Delta_{j,s}| \le (j-1)2^{\lceil n/2\rceil}. \tag{RSD}$$

Thus a Witt-precision comparison only addresses the nonresonant layers. The resonant ones occur in the target range (for example $j = 200$, $2^s = 16$). For these orders, $2^s > \ell/(3c)$. The group $2^s\mathcal{E}_j$ therefore has fewer than $8\ell^3$ elements. That fact narrows the problem, but it does not solve it. At the largest order the group is $\{1\}$, so (NSD) still contains the original identity-class term.

If we prove this bound, the rest follows:

$$\boxed{\text{HWO (or another proof of REL)}} \Longrightarrow \text{REL} \Longrightarrow (1) \Longrightarrow \text{prime in the identity class.}$$

Step 2 removes the proper powers. For every degree $n \le 3000$ a half-degree irreducible is exhibited and doubly certified—each carries a Frobenius/Bézout irreducibility certificate that an independent checker re-verifies (**FINITE**). A proof of (HWO) for $\ell \ge 200$ and $n \in \{2\ell+1, 2\ell+2\}$ would prove the conjecture for all $n$.

**Unconditional infinite families (PROVED).** Monomial substitution never leaves the window: $f(x^t) - x^{mt} = (f - x^m)(x^t)$, so $\deg(f(x^t) - x^{mt}) = t\deg(f - x^m) \le \lfloor mt/2\rfloor$ for every $t \ge 1$. Hence, by the classical criterion (Lidl–Niederreiter, *Finite Fields*, Thm. 3.35): if $f$ is an in-window irreducible of degree $m$ and order $e$, and $t \ge 2$ has every prime factor dividing $e$ but not $(2^m-1)/e$, then $f(x^t)$ is an in-window irreducible of degree $mt$, and the conjecture holds at $n = mt$. The seed $x^2+x+1$ gives $\Phi_{3^{k+1}} = x^{2\cdot 3^k} + x^{3^k} + 1$ and $n = 2\cdot 3^k$ (already noted by E. Jeřábek on MathOverflow, Nov. 2011, on the question where Lemire first posed the conjecture); the seed $x^3+x+1$ (order 7) gives the odd degrees $n = 3\cdot 7^k$; the seed $x^4+x+1$ (order 15) gives $n = 4\cdot 3^a 5^b$. Applied to the certified seeds of degree $\le 3000$ this reaches 9.3% of the composite $n \le 10^5$ (exact count, two independent engines). Every such family has density zero, and no construction of this kind ever produces a prime $n$ or a power of two: the construction route reduces the conjecture to prime degrees and stops there. The earlier claim in this note that $n = 2\cdot 3^k$ was the only provable family, giving only even $n$, was wrong. None of this touches the open estimate above.

**A second sufficient statement.** Let $K = \ker(\mathcal{E}_\ell \to \mathcal{E}_{a-1})$, an elementary abelian group of order $2^{\ell-a+1}$ (since $a > \ell/2$), and for a character $\psi$ of $K$ let

$$A_\psi = \sum_{\substack{F \text{ monic, } \deg F = n\\ \langle F\rangle_{a-1} = 1}} \Lambda(F)\,\psi(\langle F\rangle_\ell)$$

be the prime mass of the cylinder $\{x^n + g : \deg g \le n-a\}$ twisted by a sign pattern of the coefficients of $x^{n-a}, \dots, x^{n-\ell}$. Fourier inversion on $K$ gives $|N_\ell(1) - N_{a-1}(1)/|K|| \le \max_{\psi\ne 1}|A_\psi|$, and $2^\ell N_{a-1}(1)/|K| = 2^{a-1}N_{a-1}(1)$, while $B_{\ell,n} \ge 2^{2\ell-1}$ because $a - 3 < \ell \le 2^c$. Hence

$$|A_\psi| < 2^{\ell-1} \quad \text{for every } \psi \ne 1 \tag{CYL}$$

implies (REL). The Weil bound gives only $|A_\psi| \le (\ell-1)2^{\lceil n/2\rceil}$, a factor $4(\ell-1)$ short; the random model predicts $|A_\psi| \approx 2^{(n-a+1)/2}\sqrt{\ell}$. Exact computation for $14 \le \ell \le 24$ at both endpoints finds the root-mean-square of $A_\psi$ equal to that prediction within 8%, (CYL) true from $\ell = 16$ ($n$ odd) and $\ell = 18$ ($n$ even) on, with $\max_\psi |A_\psi|/2^{\ell-1}$ falling to 0.09 and 0.15 at $\ell = 24$, and the same bound true for every one of the $2^{a-1}$ translated cylinders from $\ell = 20$ on. This is finite evidence, not a proof, and it is the smallest statement currently known to close the chain.

**Why the usual bounds fail.** Taking absolute values for each character or conductor loses too much. Bounds that use only the Fourier support give no saving at all. Cauchy's inequality, the tested bent and Kerdock models, and simple Heisenberg lifts also fail. Known moment theorems over large finite fields are not uniform in this fixed-field, growing-conductor setting. We need a bound that keeps the high-order character sums together. Another table of small cases will not provide it. Nor can generic invariants of the relevant $L$-functions: for the product $P_\psi$ of the $L$-polynomials over a coset of low-conductor twists, degree, integrality, the Riemann hypothesis, the functional equation, the 2-adic Newton polygon and all power sums of degree below $n$ are preserved by multiplication with powers of $1 - 2^{(n+1)/2}T^n + 2^nT^{2n}$, which drives the $n$th power sum to about 0.7 of its trivial bound; only the specific arithmetic of the Hayes characters can supply the saving. On the integer side the best single-residue exponent for prime-power moduli is 2.1115 (Banks–Shparlinski), and asymptotics at $q^2$ are known only under GRH with a pair-correlation hypothesis.

# References


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